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Prevalence of the Prescribing of Attention-Deficit Hyperactivity Disorder Medication During Pregnancy: A Systematic Review

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ABSTRACT

Attention deficit hyperactivity disorder (ADHD) diagnosis in women has markedly increased in recent years, resulting in an increase of ADHD medication exposure during pregnancy, which has uncertain health risks. It is therefore critical to understand the extent of the issue. The objective of this novel systematic review was to investigate the prevalence of the prescribing of ADHD medication during pregnancy. A systematic review was conducted following preferred reporting items for systematic reviews and meta-analyses (PRISMA) guidelines. The last search was October 13, 2023, for articles examining the prescribing of ADHD medication in pregnancy. Fifteen studies were included. Participants from the United States ($n = 4$), the Nordic countries ($n = 10$), Canada ($n = 3$), and Australia ($n = 1$) were included. The prevalence of ADHD-medication dispensing during pregnancy ranged from 0.07 to 6.01 per 1,000 pregnancies. Typically, stimulant medications were more prevalent than nonstimulants. The most prevalent ADHD medication dispensed during pregnancy was methylphenidate followed by amphetamine-type medications. There was high heterogeneity ($I^2 = 99.95\%$) between the studies, making the study data not appropriate for meta-analysis. The prevalence of the prescribing of ADHD medications in pregnancy ranged substantially, with psychostimulant medications most prescribed. Apart from in the Nordic countries, estimates of prevalence were limited.

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Attention deficit hyperactivity disorder (ADHD); ADHD medication; pregnancy; prevalence

Attention deficit hyperactivity disorder (ADHD) is a neurodevelopmental disorder that is commonly diagnosed in children (Damer et al. 2021). It is now a widely accepted belief that the disorder persists into adulthood, affecting approximately 4%–5% of the adult population (Aoki et al. 2021). The symptoms of ADHD include hyperactivity, inattention, and impulsivity (Damer et al. 2021), which are typically associated with a deficiency in neurotransmitters including dopamine and norepinephrine (Aoki et al. 2021). While other treatments are available, pharmacotherapies are considered the first line of treatment, as they work to increase norepinephrine and dopamine neurotransmission in the prefrontal cortex (Aoki et al. 2021).

Medications commonly prescribed worldwide include stimulants such as methylphenidate and amphetamine-type stimulants (which include amphetamine, dexamphetamine, lisdexamphetamine, and dextroamphetamine), and nonstimulant medications including atomoxetine, guanfacine, modafinil and clonidine (Kooij et al. 2010; Prakash et al. 2021). Stimulant medications work by regulating dopamine and noradrenaline pathways while nonstimulants are either selective noradrenaline reuptake inhibitors or two alpha-2 agonists (Cerrillo-Urbina et al. 2018). Stimulant medications are typically more efficacious and therefore more commonly prescribed than nonstimulants (Kooij et al. 2010; Prakash et al. 2021).

Diagnosis of ADHD in women has increased in recent years in many countries (Baker et al. 2022). This is likely due to clinicians' better understanding of how ADHD presents in women, leading to more women being diagnosed as adults and prescribed ADHD medication for the treatment of their symptoms (Aoki et al. 2021; Baker and Freeman 2018). This has also led to more women of reproductive age using pharmacotherapies to treat their symptoms (Anderson et al. 2018) and a greater number of pregnancies being exposed to ADHD medications (Baker and Freeman 2018).

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There is conflicting information on the safety of ADHD medication during pregnancy (Anderson et al. 2018; Damer et al. 2021; Russell et al. 2024). Risks associated with some ADHD medications include congenital malformations (Damkier and Broe 2020), preeclampsia (Cohen et al. 2017; Poulton, Armstrong, and Nanan 2018), miscarriages (Andrade 2018; Hærvig et al. 2014), and preterm birth (Andrade 2018). In contrast, other studies have found no such risks or inconclusive evidence (Jiang et al. 2019; Li et al. 2020; Russell et al. 2024). A recent Australian study found that individuals who ceased dexamphetamine during pregnancy had higher risks of threatened abortion than individuals who continued their medication throughout their pregnancy (Russell et al. 2024). Understanding the prevalence of ADHD medication exposure in pregnancy is crucial for being informed about the potential magnitude these risks may have. Diagnosis of ADHD in females of a reproductive age has drastically increased in recent years (Madsen et al. 2024) and there is some evidence to suggest that pharmacotherapy during pregnancy may have potential risks (Andrade 2018; Cohen et al. 2017; Damkier and Broe 2020; Hærvig et al. 2014; Huybrechts et al. 2018; Poulton, Armstrong, and Nanan 2018), indicating the need to understand the extent of the issue.

There has been a growing focus on ADHD in female patients and the unique challenges associated with their diagnosis and treatment, including pregnancy. As such, there has been increasing data on the prevalence of prescribing of ADHD medication during pregnancy. Consolidating this research to understand the strengths and variability in these estimates of prevalence is an important step.

This novel systematic review aims to examine the currently available literature on the prevalence of ADHD medication prescribed during pregnancy.

Methods

Eligibility criteria

Peer-reviewed cohort studies, case-control studies, cross-sectional studies, and research letters were included. Categorizing ADHD medication prescribed during pregnancy also included discontinuing use during pregnancy. Studies were limited to those written in English and examining females taking these medications for ADHD (rather than other conditions). Expert opinions, conference abstracts, systematic reviews, scoping reviews, narrative reviews, case reports, case series, and laboratory and animal studies were excluded. Unpublished data was not sought.

Information sources

Ovid Technologies was used to search databases, which included MEDLINE, Embase, and PsycINFO. CINAHL plus was searched using EBSCOhost. These databases were last searched on October 13, 2023.

Search strategy

We followed the guidelines from the Preferred Reporting Items for Systematic and Meta-Analysis (PRISMA) statement (Page et al. 2021). The full details for the search terms and syntax used for each database is provided online in Supplementary Appendix S1. Medication for the treatment of ADHD included amphetamine-type medications (e.g., dexamphetamine, lisdexamphetamine, dextroamphetamine, and amphetamine), methylphenidate, guanfacine, modafinil, atomoxetine, and clonidine. ADHD is also known as ADD (attention deficit disorder), ADDH (attention deficit disorder with hyperactivity), ADHS (attention deficit hyperactivity syndrome), attention deficit disruptive behavior disorder, HKD (hyperkinetic disorder), and ODD (oppositional defiant disorder); these alternative names were included in the search strategy.

Selection process

Duplicates were removed initially in EndNote (The Endnote Team 2013) and then uploaded to Covidence (Veritas Health Innovation) where any missed duplicates were removed. Covidence (Veritas Health Innovation) was used to assist two reviewers with assessing the studies independently. The two reviewers independently screened the titles and abstracts and then assessed the full text of potentially eligible articles. Disagreements were solved by consensus and when consensus could not be reached a third reviewer reviewed the article. The results from the search are presented using the PRISMA flow chart (Figure 1) (Page et al. 2021).

Data collection process and data items

A template in Covidence (Veritas Health Innovation) was used to extract the data from the journal articles by one investigator (DR). Data were extracted from each study, including study ID, title, the lead author and their contact details, country in which the study was conducted, study design, database and/or registries used, start and end date of data collection, possible conflicts of interest for study authors, exposure

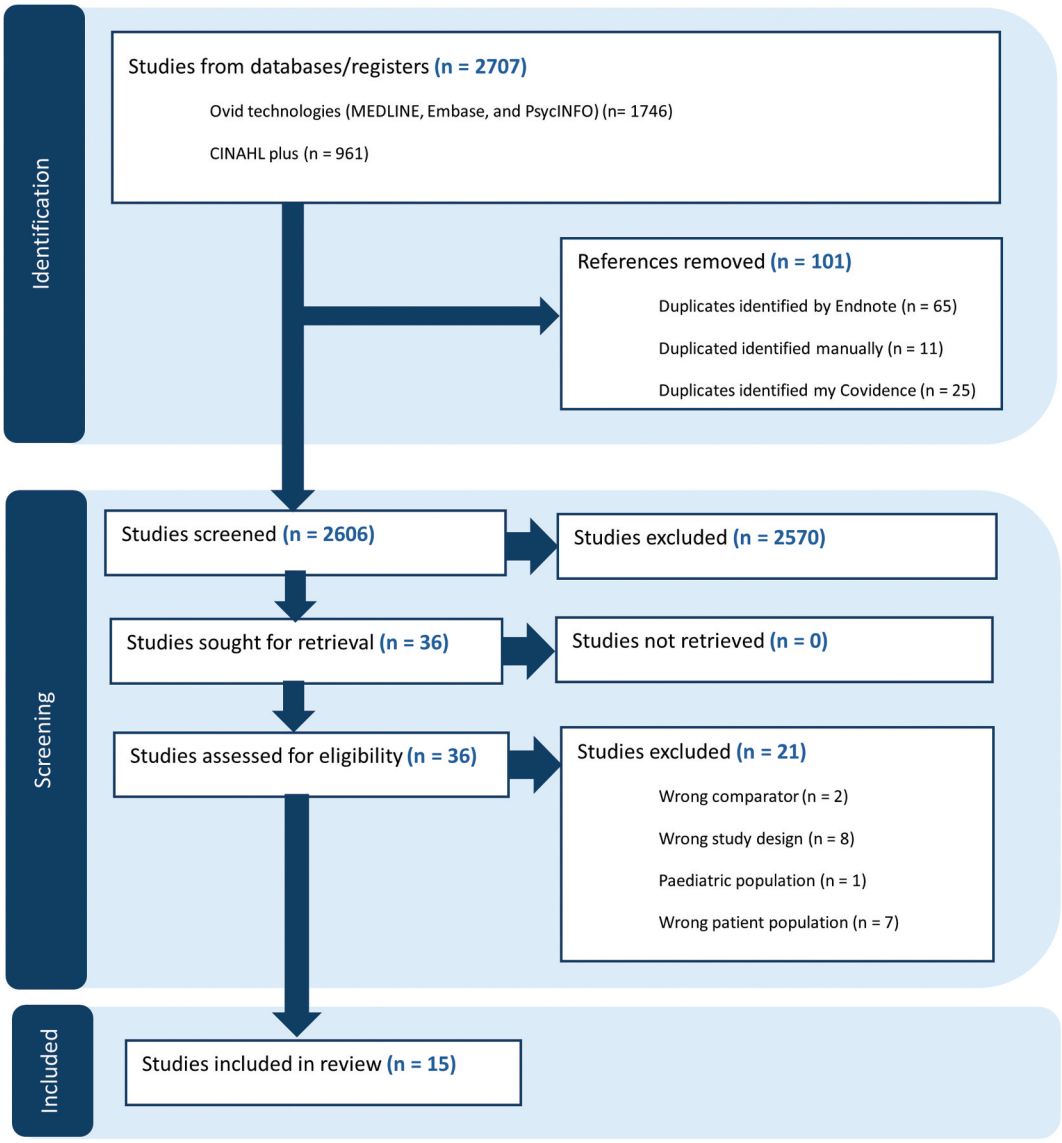


Figure 1. Preferred reporting items for systematic review and meta-analysis (PRISMA) flow chart of the studies included for the investigation of the prevalence of attention-deficit hyperactivity disorder medication during pregnancy.

definition, inclusion and exclusion criteria, types of medications included, total number of participants, total number of pregnancies exposed, number of each type of medication taken during pregnancy, number of pregnancies unexposed, and total number of pregnancies.

Study risk of bias assessment

The "Quality Assessment Checklist for Prevalence Studies" by Hoy et al. (2012) was used to determine the biases of each article by two reviewers (DR and EQ) (Hoy et al. 2012). This tool asseses both internal and external validity via the use of ten questions (Hoy et al. 2012). The overall risk for bias in each study was determined to be low when "No" was answered for 0–3 questions, moderate for when "No" was answered for 4–6 questions and high for when "No" was answered for 7 or more questions.

Synthesis methods

The study characteristics of each study were tabulated. The raw data from each study were then presented in a table with the calculated prevalence. Prevalence was calculated using Stata/SE 18.0 (StataCorp, College Station, Texas, USA 2023) by using the raw data from each individual study that was included in the final analysis of the study.

We planned to synthesize the extracted data as a meta-analysis using Stata/SE 18.0 (StataCorp, College Station, Texas, USA 2023). Due to

prevalence numbers being close to 0%, analysis was performed using Freeman-Tukey transformations (StataCorp 2023). Heterogeneity was evaluated by using I^2 statistic. As the heterogeneity was considerable (I^2 = 75%–100%) (Deeks, Higgins, and Altman 2024), it was explored by subgroup analysis. Due to this high heterogeneity, no further analysis was performed, and a qualitative summary is provided (Page et al. 2021).

Results

Study selection

A total of 2,606 potentially eligible studies were identified, after duplicates were removed (Figure 1). After screening titles and abstracts, 2,570 studies were removed, leaving 36 studies available for full-text screening. Our review included 15 published studies.

Study characteristics

The cohort studies included data from several countries: Denmark, Finland, Iceland, Norway, Sweden, the United States, Canada, and Australia (Table 1). Four studies looked at multiple regions and presented their data by country (Table 1). Nordic countries had ten cohorts (Bang Madsen et al. 2023; Bro et al. 2015; Bröms et al. 2023; Cesta et al. 2020; Cohen et al. 2023; Damkier and Broe 2020; Hærvig et al. 2014; Huybrechts et al. 2018; Kolding et al. 2021; Nörby, Winbladh, and Källén 2017), the United States had four cohorts (Bröms et al. 2023; Cohen et al. 2017; Hanley and Mintzes 2014; Huybrechts et al. 2018), Canada had three cohorts (Camacho et al. 2023; Hanley, Miller, and Mintzes 2020; Lemelin et al. 2021), and Australia had one cohort (Camacho et al. 2023) (Table 1).

The four studies examining ADHD medication prescribed in the United States utilized health-care-claim databases (Table 1). The remaining studies used data from administrative databases/registries (Table 1). Medications of interest included methylphenidate (n = 11), amphetamine (n = 7), dextroamphetamine (n = 2), dexamphetamine (n = 5), lisdexamphetamine (n = 5), modafinil (n = 6), atomoxetine (n = 10), guanfacine (n = 2), and clonidine (n = 1) (Table 1).

The period of data collection varied from study to study. The earliest collection of data was from 1996 to 2013 (Huybrechts et al. 2018); whereas, the most recent was from 2010 to 2019 (Cohen et al. 2023) (Table 1). The longest study period was for seventeen years (Bang Madsen et al. 2023; Huybrechts et al. 2018) and the shortest was for five years (Hanley and Mintzes 2014).

Five studies investigated exposure in the first trimester only (Bröms et al. 2023; Cesta et al. 2020; Damkier and Broe 2020; Huybrechts et al. 2018; Kolding et al. 2021), two addressed exposure in the first and second trimesters (Camacho et al. 2023; Cohen et al. 2017), and seven studies encompassed exposure in all three trimesters (Bang Madsen et al. 2023; Bro et al. 2015; Cohen et al. 2023; Hærvig et al. 2014; Hanley and Mintzes 2014; Hanley, Miller, and Mintzes 2020; Lemelin et al. 2021). One study had an ambiguous definition of exposure and, thus, it was not possible to ascertain the trimester exposure (Nörby, Winbladh, and Källén 2017). The studies that investigated first trimester exposure ranged in prevalence only from 0.07/1,000 pregnancies (Cesta et al. 2020) to 1.56/1,000 pregnancies (Kolding et al. 2021). The studies that investigated any exposure timeframe during pregnancy ranged from 0.19/1,000 pregnancies (Bro et al. 2015) to 6.01/1,000 pregnancies (Hanley and Mintzes 2014). All studies were deemed to have a low risk of bias.

Results of individual studies

The study size varied from 139,229 individuals in the 2023 Australian study (Camacho et al. 2023) to 2,560,069 individuals in the Nordic study (Huybrechts et al. 2018) (Table 2). Prevalence of ADHD medication prescribed in each study ranged from 0.07/1,000 pregnancies for the dispensing of modafinil in Sweden (Cesta et al. 2020) to 6.01/1,000 pregnancies for the dispensing of psychostimulants in the United States (Hanley and Mintzes 2014) (Figure 2). A study conducted in the United States had the highest prevalence of ADHD medication dispensed during pregnancy at 6.01/1,000 pregnancies (Hanley and Mintzes 2014) (Table 3) (see Supplementary Appendix S2, Figure S3). Stimulants were more commonly dispensed during pregnancy than nonstimulants, 0.37/1,000–6.01/1,000 pregnancies and 0.02/1,000–0.41/1,000 pregnancies, respectively (Table 3) (see Supplementary Appendix S2, Figure S4). The stimulant that had the highest prevalence during pregnancy was methylphenidate, ranging from 0.17/1,000 to 3.51/1,000 pregnancies, while amphetamine-type medications ranged from 0.02/1,000 to 3.07/1,000 pregnancies (Table 3) (see Supplementary Appendix S2, Figure S5).

Results of syntheses

The overall pooled prevalence of ADHD medication prescribed in pregnancy was 1.49/1,000 pregnant persons (95%CI: 0.80, 2.40) with I^2 determining considerable heterogeneity (I^2 = 99.95%) (Figure 2).

Table 1. Overview of studies included in a systematic review.

Study	Location	Study period	Database/registries	Study design	Medications
Bang Madsen et al. (2023)	Denmark	1998–2015	Danish Medical Birth Registry; Danish National Prescription Registry; Danish Medical Birth Register; Danish Psychiatric Central Research Registry; Danish National Prescription Register; Danish National Patient Register; Statistics Denmark's registers on socioeconomic status	Cohort study	Methylphenidate, amphetamine, dexamphetamine, lisdexamphetamine, modafinil, atomoxetine, and clonidine
Bröms et al. (2023)	United States, Denmark, Finland, Iceland, Norway, and Sweden	2001–2017	U.S. Medicaid Analytic eXtract (MAX); the national health registers of each Nordic country	Cohort study	Atomoxetine
Bro et al. (2015)	Denmark	1997–2008	Danish Medical Birth Registry (MBR); the Danish National Hospital Discharge Register (NHDR); Registry of Medicinal Product Statistics; Danish Psychiatric Central Registry (DPCR)	Cohort study	Methylphenidate and atomoxetine
Camacho et al. (2023)	Canada and Australia	2002–2017	ICES20 (Ontario); MOMBABY database; Narcotics Monitoring System; Ontario Drug Benefit database; the Maternal Use of Medications and Safety (MUMS); Perinatal Data Collection	Cohort study	Methylphenidate, amphetamine, dexamphetamine, lisdexamphetamine, and atomoxetine
Cesta et al. (2020)	Norway and Sweden	2005–2017	Nationwide medical birth registers in Norway and Sweden; nationwide prescribed drug registers in Norway and Sweden; national patient registers in Norway and Sweden	Cohort study	Modafinil
Cohen et al. (2017)	United States	2000–2010	Medicaid Analytic eXtract (MAX) database	Cohort study	Amphetamine, dextroamphetamine, methylphenidate, and atomoxetine
Cohen et al. (2023)	Norway and Sweden	2010–2019	Medical Birth Registry of Norway (MBRN); Norwegian Prescription Database (NorPD); Swedish Medical Birth Register (SMBR); Prescribed Drug Register (PDR)	Cohort study	Amphetamine, dexamphetamine, methylphenidate, atomoxetine, lisdexamphetamine, and guanfacine
Damkier and Broe (2020)	Denmark	2004–2017	Danish national health registries	Cohort study	Methylphenidate and modafinil
Hærvig et al. (2014)	Denmark	1998–2010	Medical Birth Registry; National Patient Registry; the Register of Legal Induced Abortion; National Prescription Registry; Psychiatric Central Research Registry; Statistics Denmark	Cohort study	Methylphenidate, atomoxetine, and modafinil
Hanley, Miller, and Mintzes (2020)	Canada	1997–2010	Vital statistics birth records; BC's Perinatal Data registry; medical services records; hospital discharge records; BC PharmaNet	Cohort study	Psychostimulants
Hanley and Mintzes (2014)	United States	2006–2011	Truven Health MarketScan database	Cohort study	Psychostimulants
Huybrechts et al. (2018)	United States, Denmark, Finland, Iceland, Norway, and Sweden	1996–2013	Medicaid Analytic eXtract (MAX); nationwide health registries in the Nordic countries	Cohort study	Methylphenidate, amphetamine, and dextroamphetamine
Kolding et al. (2021)	Denmark	2007–2014	Danish Fetal Medicine Database; the Danish National Patient Registry; the Danish Medical Birth Registry; the Danish Health Services Prescription Database	Cohort study	Methylphenidate, modafinil, and atomoxetine
Lemelin et al. (2021)	Canada	1998–2015	Quebec Pregnancy/Children Cohort; Régie de l'Assurance Maladie du Québec (RAMQ); MedEcho, Institut de la Statistique du Québec (ISQ); the Ministère de l'Éducation et de l'Enseignement Supérieur du Québec (MEES); Quebec Public Prescription Drug Insurance Database; Quebec Statistics Database	Cohort study	Methylphenidate, amphetamine mixed salts, lisdexamphetamine, dexamphetamine, atomoxetine, and guanfacine
Nörby, Winbladh, and Källén (2017)	Sweden	2006–2014	Medical Birth Register (MBR); Prescribed Drug Register (PDR); Swedish Neonatal Quality Register (SNQ); Perinatal Revision South Register (PRS)	Cohort study	Methylphenidate, amphetamine, dexamphetamine, lisdexamphetamine, modafinil, and atomoxetine

Heterogeneity was explored using subgroup analysis. This did not show a difference between regions (chi-squared test statistic: 6.22; $p = .10$) and thus did not substantially reduce heterogeneity when different regions were analyzed (see Supplementary Appendix S2, Figure S3).

Further subgroup analysis was completed by looking at stimulants versus nonstimulants, which showed

Table 2. Prevalence of attention-deficit hyperactivity disorder medication exposure during pregnancy by region.

Study	Region	Total number of participants	Total number exposed	Number of unexposed	Prevalence per 1,000 pregnancies
Bang Madsen et al. (2023)	Denmark	1,068,073	898	1065905	0.84
Bröms et al. (2023)	Denmark, Finland, Iceland, Norway, and Sweden	2,440,606	368	–	0.15
	United States	1,797,938	622	–	0.35
Bro et al. (2015)	Denmark	989,932	186	989746	0.19
Camacho et al. (2023)	Canada	554,272	1,360	552912	2.45
	Australia	139,229	146	139083	1.05
Cesta et al. (2020)	Norway and Sweden	1,917,605	133	1917472	0.07
Cohen et al. (2017)	United States	1,466,792	5,299	1461493	3.61
Cohen et al. (2023)	Norway	577,116	2,339	574777	4.05
	Sweden	1,118,988	5,436	1113554	4.86
Damkier and Broe (2020)	Denmark	829,656	1,012	828644	1.22
Hærvig et al. (2014)	Denmark	1,054,494	480	1054014	0.46
Hanley, Miller, and Mintzes (2020)	Canada	424,307	268	–	0.63
Hanley and Mintzes (2014)	United States	343,299	2,062	–	6.01
Huybrechts et al. (2018)	United States	1,813,894	7,643	1797938	4.21
	Denmark, Finland, Iceland, Norway, and Sweden	2,560,069	1,402	2557001	0.55
Kolding et al. (2021)	Denmark	364,012	569	363219	1.56
Lemelin et al. (2021)	Canada	428,505	1,130	427375	2.64
Nörby, Winbladh, and Källén (2017)	Sweden	964,734	1,591	963143	1.65

a significant difference between the different types of medication (chi-squared test statistic: 24.95; $p < .01$) (see Supplementary Appendix S2, Figure S4), still heterogeneity remained high for stimulants ($I^2 = 99.94\%$), non-stimulants ($I^2 = 99.36\%$), and mixed medications ($I^2 = 99.93\%$). This was also seen when the different specific medications were analyzed (chi-squared test statistic: 90.42; $p < .01$) (see Supplementary Appendix S2, Figure S5). The heterogeneity was not significantly reduced for amphetamine-type medications ($I^2 = 99.95\%$), atomoxetine ($I^2 = 99.49\%$), methylphenidate ($I^2 = 99.88\%$), and mixed medication ($I^2 = 99.94\%$). Heterogeneity was reduced for modafinil ($I^2 = 80.39\%$); however, this included only three studies and therefore needs to be interpreted with caution. Only one study investigated guanfacine and therefore the reduction in heterogeneity is not comparable. Due to the considerable heterogeneity, no further analysis was completed in line with PRISMA guidelines (Page et al. 2021).

Summary of papers

All but one paper included in this review were completed at the level of the pregnancy, meaning that an individual could be included multiple times. Hanley and Mintzes (2014) restricted their investigation to the first pregnancy when more than one pregnancy was identified. The papers were evenly divided by those who looked at all births (including live births, still births, miscarriages, and pregnancy terminations) ($n = 7$) and by those who looked only at live births ($n = 8$). They were also evenly divided into those who looked at singleton ($n = 8$) and multiple ($n = 7$) pregnancies.

Studies commonly excluded pregnancies exposed to known teratogens (Camacho et al. 2023; Damkier and Broe 2020; Huybrechts et al. 2018), where gestational age was unknown, an implausible data value was recorded (Bang Madsen et al. 2023; Cohen et al. 2017; Damkier and Broe 2020), or there were any chromosomal abnormalities (Cesta et al. 2020; Damkier and Broe 2020; Huybrechts et al. 2018). Cohen et al. (2017) also excluded pregnancies resulting in neonates with congenital anomalies. Congenital anomalies were however a key outcome of interest for many studies ($n = 9$, 60%) which investigated the association between prenatal exposure to ADHD medications and congenital anomalies (Bro et al. 2015; Bröms et al. 2023; Camacho et al. 2023; Cesta et al. 2020; Damkier and Broe 2020; Hærvig et al. 2014; Kolding et al. 2021; Nörby, Winbladh, and Källén 2017). Camacho et al. (2023) excluded infants with extreme birth weight in relation to gestational age, while Bang Madsen et al. (2023) excluded children with missing paternal links.

All of the studies except for one (Nörby, Winbladh, and Källén 2017) used dispensed prescriptions to ascertain exposure of the ADHD medication during pregnancy. Nörby, Winbladh, and Källén (2017) used self-reporting of taking the medication during pregnancy and/or if the medication was registered in a prescription database during the pregnancy timeframe or one month prior to pregnancy. Ten studies defined exposure by requiring at least one prescription during

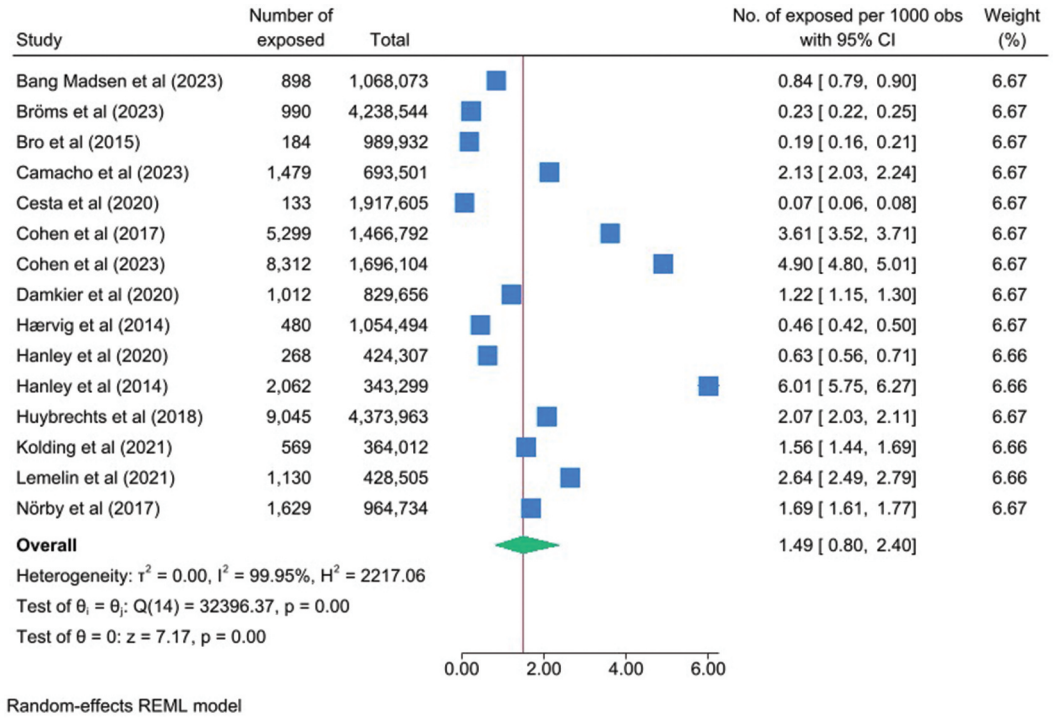


Figure 2. Forest plot of prevalence of exposure during pregnancy to attention-deficit-hyperactivity-disorder medication.

Table 3. Prevalence of attention-deficit hyperactivity disorder medication exposure during pregnancy by medication type.

Study	Medication	Total number of participants	Number of exposed	Number of unexposed	Prevalence per 1,000 pregnancies
Bang Madsen et al. (2023)	Methylphenidate	1,068,073	703	1065905	0.658
	Amphetamine-type		20		0.019
	Atomoxetine		125		0.117
	Unspecified		50		0.047
Bröms et al. (2023)	Atomoxetine	4,238,544	990	–	0.234
Bro et al. (2015)	Methylphenidate	989,932	166	989746	0.168
	Atomoxetine		18		0.018
	Both methylphenidate & atomoxetine		2		0.002
Camacho et al. (2023)	Methylphenidate	693,501	Canada = 749 Australia = 22–27	691995	Canada = 1.080 Australia = n/a
	Amphetamine-type		696		1.004
	More than one stimulant		Canada = 34 Australia ≤ 5		Canada = 0.049 Australia = n/a
Cesta et al. (2020)	Modafinil	1,917,605	133	1917472	0.069
Cohen et al. (2017)	Methylphenidate	1,466,792	1,515	1461493	1.033
	Amphetamine-type		3,331		2.271
	Atomoxetine		453		0.309
Cohen et al. (2023)	Methylphenidate	1,696,104	2,339	1688331	1.379
	Amphetamines-type		5,436		3.205
	Atomoxetine		677		0.399
	Guanfacine		Norway <3 Sweden = 14		Norway = n/a Sweden = 0.008
Damkier and Broe (2020)	Methylphenidate	829,656	963	828644	1.161
	Modafinil		49		0.059
Hærvig et al. (2014)	Methylphenidate	1,054,494	81.88%	1054014	0.819
	Atomoxetine		9.38%		0.094
	Modafinil		8.75%		0.087
Hanley, Miller, and Mintzes (2020)	Stimulants	424,307	268	–	0.632
Hanley and Mintzes (2014)	Stimulants	343,299	2,062	–	6.006
Huybrechts et al. (2018)	Methylphenidate	4,373,963	3,474	4354939	0.794
	Amphetamine-type		5,571		1.274
Kolding et al. (2021)	Methylphenidate	364,012	473	363219	1.299
	Unspecified		69		0.189
Lemelin et al. (2021)	Unspecified	428,505	1,130	427375	2.637
Nörby, Winbladh, and Källén (2017)	Atomoxetine	964,734	165	963143	0.171
	Stimulant		1,464		1.517

the pregnancy period of interest (Bang Madsen et al. 2023; Camacho et al. 2023; Cesta et al. 2020; Cohen et al. 2017, 2023; Damkier and Broe 2020; Hanley and Mintzes 2014; Hanley, Miller, and Mintzes 2020; Huybrechts et al. 2018; Lemelin et al. 2021). The other four studies did not specify the number of scripts required to define the pregnancy as exposed (Bro et al. 2015; Bröms et al. 2023; Hærvig et al. 2014; Kolding et al. 2021).

Out of the fifteen papers reviewed, only two listed their primary aims as investigating the prevalence of ADHD medication exposure during pregnancy (Cohen et al. 2023; Lemelin et al. 2021). Two studies were conducted to investigate the patterns and trends of ADHD medication dispensed during pregnancy (Hanley and Mintzes 2014; Hanley, Miller, and Mintzes 2020). The majority of the papers investigated health outcomes associated with the pregnancy, the mother, or/and the neonate (Bro et al. 2015; Bröms et al. 2023; Camacho et al. 2023; Cesta et al. 2020; Cohen et al. 2017; Damkier and Broe 2020; Hærvig et al. 2014; Huybrechts et al. 2018; Kolding et al. 2021; Nörby, Winbladh, and Källén 2017). One study investigated long-term health outcomes of the children (Bang Madsen et al. 2023).

Discussion

In this review, the prevalence of ADHD medication prescription during pregnancy ranged from 0.07/1,000-–6.01/1,000 pregnancies. Only fifteen papers were examined which investigated data from the United States, Canada, Australia, and the Nordic countries. Medications that were explored included methylphenidate, amphetamine-type stimulants (e.g., amphetamine, dextroamphetamine, dexamphetamine, and lisdexamphetamine), modafinil, atomoxetine, guanfacine, and clonidine.

There was considerable heterogeneity seen between the studies, making an estimate of the overall prevalence inappropriate. The possible causes of this heterogeneity could be due to the types of ADHD medications included in the studies, whether the studies considered only one medication or multiple medications, a difference in study definitions, and/or a difference between the country the studies were conducted in. Even seemingly similar countries can have substantial differences in regulations for ADHD medications, the availability of medications, methods for diagnosing ADHD, health-care practices, and data collection methods. Similarly, there are likely cultural differences in the perceptions of ADHD between countries. However, this needs to be interpreted with care due to the limited number of studies available for some medications. Psychostimulants, such as methylphenidate and amphetamine-type medications, are typically more commonly dispensed than nonstimulants (atomoxetine, modafinil, guanfacine and clonidine) (Cortese et al. 2018). This is reflected in the prevalence patterns seen between stimulants (0.17/1,000–6.01/1,000 pregnancies) and nonstimulants (0.02/1,000–0.41/1,000 pregnancies) in our results.

The prescribing of modafinil and atomoxetine was lower and more homogenous in prevalence, this was especially seen in modafinil. These medications are both nonstimulants and are typically considered second-line treatments (Cortese et al. 2018). Efficacy and therefore potential patient uptake are typically lower for modafinil than that of a stimulant medication, which could potentially contribute to the lower uptake compared with stimulant medications (Cortese et al. 2018; Heal, Cheetham, and Smith 2009). In adults, atomoxetine is typically not as efficacious as stimulant medications, which is potentially why the decrease in uptake was not as low as what was seen for modafinil (Cortese et al. 2018). The tolerability of these medications would also affect uptake (Cortese et al. 2018).

Another possible cause of heterogeneity could be related to the years in which the data were collected. The majority of the studies imply that there was an increasing number of females of a reproductive age being diagnosed with ADHD and thus treated with medication (Bang Madsen et al. 2023; Bro et al. 2015; Bröms et al. 2023; Camacho et al. 2023; Cohen et al. 2017, 2023; Hærvig et al. 2014; Hanley and Mintzes 2014; Hanley, Miller, and Mintzes 2020; Huybrechts et al. 2018; Kolding et al. 2021; Lemelin et al. 2021; Nörby, Winbladh, and Källén 2017). This would mean that studies completed earlier would have a lower prevalence than more-recent studies, contributing to the difference that is shown in the analysis. This would be difficult to examine using the available data, due to the variation in length of data collection and the small number of studies.

Examination of prevalence by timing of exposure was typically limited to early pregnancy, with estimates of second and third trimester exposure not estimated in any study. Typically, the use of medications for pre-existing conditions (e.g., conditions not arising during pregnancy) declines following the first trimesters. This often aligns with people becoming aware of their pregnancy and choosing to cease their medication to reduce the potential harms associated with prenatal exposure (Kelty et al. 2024; Lemelin et al. 2021; Srinivas et al. 2023). However, further examination of prenatal exposure to ADHD medications by trimester is required.

Interestingly, there were no results from low and middle income countries, therefore it is difficult to ascertain the prevalence of ADHD medication exposure during pregnancy in these regions. Further investigations in these countries would provide more-robust evidence for worldwide prevalence of ADHD medications during pregnancy.

Limitations

One limitation to this review is the small number of studies available, limiting the statistical power. This also made it difficult to analyze bias, as such tests require at least ten studies per outcome to be accurate (StataCorp 2023). As all of these papers were cohort studies, they will inherently have similar biases that are commonly associated with cohort studies. They are only able to investigate dispensed or prescribed medication, which may not be an accurate reflection of what the pregnant person actually consumed. Another limitation to this study was the omission of evidence from gray literature and unpublished articles. High levels of heterogeneity made it inappropriate to perform metanalysis and thus made it impossible to provide an accurate generalized prevalence of ADHD medication use during pregnancy. Another limitation stems from the fact that the study included only articles written in English, as this may have potentially excluded some supportive literature and induced language bias; however, this is unlikely to have altered results as only about 5% of papers are in languages other than English (Pieper and Puljak 2021).

Future research

While there has been an increasing interest in the safety and prevalence of use of ADHD medications in pregnancy, research thus far shows a high amount of variability in estimates of prevalence. Ideally, future research would seek to look outside North American and Nordic countries, particularly into low- and middle-income countries. Additionally, future research should consider including estimates by the timing of exposure, overall and medication estimates, and estimates by patient demographics.

Conclusion

The prescribing of ADHD medications during pregnancy is associated with substantial variation between countries, ranging from 0.07 per 1,000 pregnancies in Nordic countries to 6.01 per 1,000 pregnancies in the United States. This study includes a limited number of studies and thus there is a substantial gap in the literature on ADHD medication during pregnancy, especially from low- and middle-income countries. Given the reported growth in the use of ADHD medications in pregnancy and the potential risks associated with prenatal exposure, understanding the prevalence of ADHD medication use in pregnancy and the health ramifications requires comprehensive research efforts globally.

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Availability of data, code, and other materials

Data and code may be provided on reasonable request.

Registration and protocol

This systematic review was registered with Open Science Framework on May 2, 2023. The registration DOI is https://doi.org/10.17605/OSF.IO/CHXE2.

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